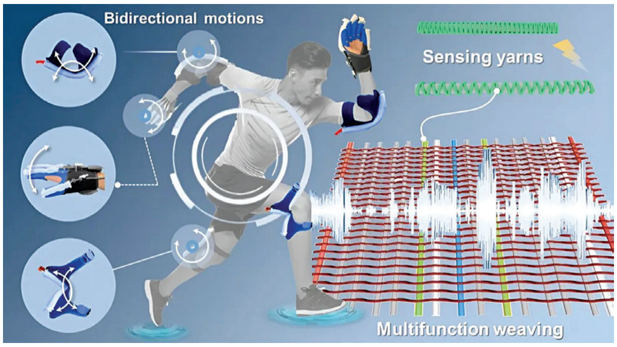
Customisable woven actuators

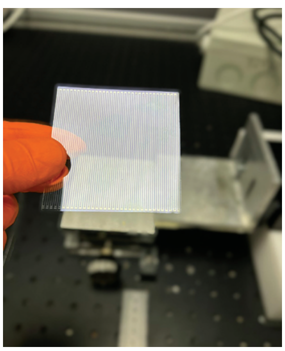
Researchers from Jiangnan University have developed a scalable and customisable textile engineering method to create woven, soft actuators for healthcare technologies and robotics. Their approach organises warp and weft yarns during the weaving process to precisely customise actuator properties. When stretched, the actuators' conductive yarns separate, disrupting electrical signals, enabling strain detection. An advantage is that these sensing yarns are fully integrated into fabrics, maintaining flexibility without adding weight or stiffness. The method also enables multi-morphing actuators capable of bending, twisting, and spiralling, which is ideal for robotic grippers mimicking animal movements like octopus tentacles.



Schematic representation of soft weaving robots for healthcare wearable devices (Source: Li et al)

Triboelectric nanogenerators produce power from movement

Researchers at the University of Surrey's Advanced Technology Institute have developed an efficient nanotechnology that converts everyday movements, like running, into electrical power for wearable devices. These flexible nanogenerators have shown a 140-fold increase in power density compared to traditional ones, generating over 1000 milliwatts of power. The technology uses a triboelectric effect, where materials become electrically charged through contact and separation. This could revolutionise energy harvesting for small devices, such as self-powered healthcare sensors. The team plans to commercialise the technology, highlighting its potential in green energy solutions for the growing number of IoT devices.



The technology aims to drive new activities in sustainable health technology, with an emphasis on industrial scalability (Credit: University of Surrey)

Manganese as alternative for lithium-ion batteries

Researchers are investigating manganese (Mn) as a sustainable, cost-effective alternative for Li-ion battery production. A study published in ACS Central Science on 26th August explores using a lithium/manganese-based material for the positive electrode, aiming to match the performance of nickel/cobalt-based materials while reducing costs. The team, led by Naoaki Yabuuchi, synthesised nanostructured $LiMnO_2$ with a monoclinic layered structure, enabling a transition to a high-performance spinel-like phase. This innovation achieved energy densities of 820Wh kg^{-1}, outperforming nickel-based materials. However, challenges like manganese dissolution remain, though solutions like lithium phosphate coatings show promise for future EV applications.

Drone-based electric grid monitoring for remote areas

Researchers at Oak Ridge National Laboratory (ORNL) have developed a drone-based system to improve electric grid monitoring, especially in remote areas. The system, demonstrated with EPB of Chattanooga, uses drones equipped with sensors to detect grid anomalies like electrical arcing. During a recent test, a scout drone was triggered by a simulated electrical fault, relaying real-time data to EPB and ORNL's labs. This approach enables faster, more efficient inspections without human presence, reducing costs and preventing outages.

ORNL researchers demonstrated the use of drones equipped with cameras and other sensors to check power lines at an EPB of Chattanooga training centre for electrical line workers (Credit: Jason Richards/ORNL, US Dept. of Energy)

Test suite for enhancing PUF security

Crypto Quantique has introduced TuRiNG, a randomness test suite specifically designed for physical unclonable functions (PUFs). PUFs are crucial in semiconductors for secure random number generation, device identities, and cryptographic keys. Unlike traditional RNGs, PUFs generate fixed-length outputs, making existing randomness tests inadequate. TuRiNG adapts tests from NIST 800-22 and includes a specialised test to ensure the independence of PUF outputs, vital for cryptographic security. The suite is comprehensive yet efficient, minimising data requirements and accounting for spatial correlations in PUFs.